Course Handout

Institute/School Name	Chitkara University Institute of Engineering and Technology		
Department Name	Department of Computer Science & Engineering		
Programme Name	Bachelor of Engineering (B.E.), Computer Science & Engineering		
Course Name	Problem Solving using Python Programming	Session	July 2024- Dec-2024
Course Code	24CSE0101	Semester/Batch	1st /2024
L-T-P(Per Week)	3-0-4	Course Credits	05
Pre-requisite	NA	NHEQF Level[1]	4.5
Course Coordinator	Dr. Mandeep Kaur	SDG Number[4]	4, 9

1. Objectives of the Course

Python is an open-source, high-level, dynamically-typed, portable, expressive, easy to learn, and well known programming language. Python is available as an open-source offerring and is associated with large global community support. Top companies such as Google, Yahoo!, Pinterest, Disney, Nokia, IBM have python as one of their main programming language. This course provides a wide scope of learning & understanding of python programming . The main objectives of the course are:

1. To understand the basics of Python (operations, control structures, data types, etc.)
2. Apply various data structures (list, tuple, string, dict.) for problem solving, fostering Industry, and Innovation.
3. Design and implement logic-building programs using Python constructs to enhance Quality Education.
4. Understand the object-oriented program design and development.
5. Understand class inheritance and polymorphism for advanced programming skills, aligned with Quality Education.
6. Design and develop real-time applications using Python GUI programming, promoting Industry, Innovation, and Infrastructure.

2. Course Learning Outcomes (CLOs)

Student should be able to:

	CLOs	Program Outcomes (PO)	NHEQF Level Descriptor[2]	No. of Lectures
CLO01	Understand python lexical features, structures and flow control for logic building.	PO1,PO3,PO10	Q1, Q2	**12**
CLO02	Apply decision statements and loops in python to solve complex problems.	PO1,PO2,PO3,PO4	Q3, Q6	**14**
CLO03	Implement Python functions to facilitate code reuse and manipulate strings.	PO3,PO5,PO12	Q3	**16**
CLO04	Create and manipulate structured data using lists, sets, tuples, and dictionaries to develop real-world solutions.	PO2,PO3,PO5,PO9,PO11,PO12	Q1, Q2	**16**
CLO05	Demonstrate the use of built-in functions to navigate the file system and manage data efficiently.	PO3,PO4,PO5,PO7,PO10,PO11,PO12	Q3	**18**
CLO06 (Only for lab components)	Develop real-world applications using the Object-Oriented Approach and Python GUI programming.	PO3,PO5,PO7,PO10,PO11,PO12	Q3, Q6	**14**
Total Contact Hours				**90**

CLO-PO Mapping

CLO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	Type of Assessment's[3]
CLO01	H		H							L			Formative,Summative
CLO02	H	H	H	M									Formative,Summative
CLO03			H		L							M	Formative,Summative
CLO04		H	H		M				M		H	M	Formative,Summative
CLO05			M		H		H			H	M	L	Formative,Summative
CLO06		H	H		M						H	M	Formative

H=High, M=Medium, L=Low

3. Recommended Books:

B01: Martin C. Brown., "The Complete Reference", MacGraw Hill Education 2021.

B02: Dr. R. Nageswara Rao, "Core Python Programming", Dreamtech Press 2019.

B03: Thareja, Reema, "Python Programming: Using Problem Solving Approach". Oxford UniversityPress c2017.

B04: Ravichandran, R.Saravanan, "Learn to Master Python", Staredu Solution, c2018.

B05: Paul Barry, "Head First Python",

B06: Roberto Tamassia, Michael H Goldwasser, Michael T Goodrich, "Data Structures andAlgorithms in Python", 1st Edition, Wiley India Pvt Ltd, 2016. ISBN-13: 978-8126562176

4. Other readings and relevant websites:

SerialNo	Link of Journals, Magazines, websites and Research Papers
1.	https://www.python.org/
2.	https://learnpython.org/
3.	https://www.codecademy.com/learn/python
4.	http://www.pyschools.com/
5.	https://www.codementor.io/learn-python-online

5. Recommended Tools and Platforms

Python 3 IDLE, Anaconda 3 (Jupyter notebook 3.6.0), Google colab, Replit

6. Course Plan: Theory+ Lab Plan

Theory Plan

Lect. No.	Topic(s)
1-2	Introduction to Python programming: Python environment setup, identifiers, reserved words ,lines and indentation, multiline statements, comments, quotation, input/output, output formatting(Integers and string). Fundamentals: Variables(simple, assigning multiple values, output variables, global/local variables)
3-8	Python decision making: If else and Nested, If else and elif. Match statement, Introductionto while and for loop.
9-14	Concept of nested looping. For loop and range method, Break, Continue and Pass, PatternDesign.
15-18	Defining and calling Functions, Type of arguments (required, keyword, default, variable length), Scope Rules(Global/local Statements), Passing Arguments(call by value/call by reference), Recursive Functions, Lambda Expressions.
19-22	Lists: Introduction to Lists, Operations on Lists, Storage Structure, Resizing and Looping on Lists, Indexing and Slicing, Passing variables through Functions, passing Lists throughFunctions.
ST1 (Syllabus=30% (Lect. No. 1-22))	
23-24	List Comprehension, map, filter & reduce functions, Two Dimensional Lists: Introduction andStorage, Jagged Lists.
25-26	String- Introduction, methods, comprehension and its relative properties.
27-28	Tuples, Set and Dictionaries- Introduction, methods, comprehension and its relativeproperties.
ST2 (Syllabus=60% (Lect No. 1-28))	
29-32	Modules: Creating Modules, Variables in Modules, Imports and Attributes, Namespaces ,Reloading, Built in modules, Generating Random values.

33-36	Files and Directories, File I/O, File positioning, File operators.
37-40	Exception Handling: Introduction to Exception, Exception Handling, Except clause, Try,finally clause, User Defined Exceptions.
ST3 (Syllabus=80% (Lect No. 1-40))	
41-45	Exploring Object-Oriented Programming In Python-Introduction, Creating classes, objects, attributes. Constructors., Idea of inheritance, Superclass And SubClass. Overriding methods,Object class , Isinstance function.
End Term Exam (Syllabus=100% (Lect No. 1-45))	

Lab Plan

Lab No.	Topic(s)
1-2	Anaconda Installation, Running Python Programs, Writing Python Scripts with Jupyter Notebook, and also make familiar with various platforms used for python programming, Implementation of basic concepts of python
3-8	Data types (text type, numeric, sequence, mapping, set), Python basic operators-arithmetic, comparison, assignment operators, logical operators. , mix type conversion, integer to float and float to integer conversion.
9-14	Coding problems related to conditional statements (if, elif, nested if else) and match statement.
15-18	Coding problems related to for/while loops, For loop and range method, Break,continue and pass. Nested loops and pattern designs
19-22	Implementation related to Functions: User Defined and Built in functions(Python mathematical functions, random number functions), Passing Arguments(call byvalue/call by reference), Recursive Functions, Lambda Expressions.
23-24	Implementation of lists. Creating empty list, Initializing list, list indexing andslicing operations, input method, Concatenating multiple lists, generating range based lists. list based methods like append(), extend(), insert(), index(), count(),sort(), reverse(), pop() and nested lists.
25-26	Implementation of 2-D list with basic operations.
27-28	Python strings- accessing values in strings, updating strings, string special characters, string formatting operators, triple quotes, built in string methods. Capitalize(), count(), endswith, startswith, find, index, isalpha, isnum, isdigit, islower, isnumeric, isupper, lower, max, min, replace, splitswith, strip.
29	Implementation of Sets and relative methods.
30	Implementation of Tuples and relative methods.
31-32	Creating empty dictionaries, initializing dictionaries, Accessing dictionary items,merging, and deletion. Understanding dictionary specific methods, keys (), values (), items (), copy (), update (), pop () and dictionary comprehension, Python dateand time functions.
33-34	Python modules-import, from… import statements, dir function, GeneratingRandom values (random module)
35	File handling- Writing to a file, Reading from a file, Resetting the file currentposition, fseek
36-37	Exploring Object-Oriented Programming In Python:, Creating classes, objects,attributes (implementation with Examples)
38	Constructors and idea of inheritance., Superclass And SubClass (implementationwith Examples)
39	Overriding methods, Object class , Isinstance function, Exploring Operator Overloading- Indexing And Slicing, Iteratable Objects (implementation with Examples)
40-41	Introduction to GUI Programming (Tkinter), Creating user defined modules, importing existing modules, Creating root windows and calling various dialogboxes on button click response.
42	Adding a Widget, Working with widgets(Displaying Text and Images With Label Widgets, Displaying Clickable Buttons With Button Widgets, Getting User Input With Entry Widgets, Getting Multiline User Input With Text Widgets, Assigning Widgets to Frames With Frame Widgets, Naming Conventions)
43	Controlling Layout With Geometry Managers(.pack(), .place(), .grid()), Using eventand event handlers(.bind())
44	Dealing with text-boxes, list-boxes, menus and dealing with basic events.
45	Building a simple App using Tkinter, Connecting Python With Oracle 11G- Performing insertion,Updation and

	deletion operation using python GUI.
	Continuous Evaluation

7. Delivery/Instructional Resources Theory Plan:

Lect. No.	Topics	CLO	Book No, CH No, Page No	TLM	ALM	Web References	Audio-Video
1-2	Introduction to Python programming. Python environment setup, identifiers, reserved words ,lines and indentation, multiline statements, comments, quotation, input/output, output formatting(Integers and string). Fundamentals: Variables(simple, assigning multiple values, output variables, global/local variables)	CLO01	B01, CH 1, Page no 1-17 B02, CH 2, Page No 19-24 B03, CH 03, Page No 83-124	Lecture	Quiz, Test questions	https://www.anaconda.com/ https://www.learnpython.org/ https://docs.python.org/3/tutorial/introduction.html	https://nptel.ac.in/courses/106106182
3-8	Python decision making- If else and Nested, If else and elif. Match statement, Introduction to while and for loop.	CLO02	B02, CH 6, Page No 117-149 B03, CH 04, Page No 137-179	Lecture	Quiz, Puzzles, Tests	https://docs.python.org/3/tutorial/controlflow.html	https://nptel.ac.in/courses/106106182
9-14	Concept of nested looping. For loop and range method, Break, Continue and Pass, Pattern Design.	CLO02	B02, CH 6, Page No 117-149	Lecture	Quiz, Brain Storming Sessions	https://docs.python.org/3/tutorial/controlflow.html https://docs.python.org/3/tutorial/controlflow.html#for-statements	https://nptel.ac.in/courses/106106182 https://nptel.ac.in/courses/106106182
15-18	Defining and calling Functions, Type of arguments (required, keyword, default, variable length), Scope Rules(Global/local Statements), Passing Arguments(call by value/call by reference), Recursive Functions, Lambda Expressions.	CLO03	B01, CH 4, Page No 81-98 B02, CH 9, Page No 237-280	Lecture	Quiz, Peer Review	https://docs.python.org/3/tutorial/controlflow.html#more-on-defining-functions	https://www.coursera.org/lecture/python-genomics/lecture-5-1-functions- part-1-5-54-PGXxZ https://www.coursera.org/lecture/python-genomics/lecture-5-2-functions- part-2-8-20-ylbSK
19-22	Lists: Introduction to Lists, Operations on Lists, Storage Structure, Resizing and Looping on Lists, Indexing and Slicing,Passing variables through Functions, passing Lists through Functions.	CLO04	B02, CH 10, Page No 283-306	Lecture	Test Questions, Quiz	https://docs.python.org/3/tutorial/datastructures.html	https://nptel.ac.in/courses/106106182
23-24	List Comprehension, map, filter & reduce functions, Two Dimensional Lists: Introduction and Storage, Jagged Lists.	CLO04	B02, CH 10, Page No 283-306	Lecture	Test Questions	https://docs.python.org/3/tutorial/datastructures.html#list-comprehensions	https://nptel.ac.in/courses/106106182

			B04, CH 08, Page No 238				
25-26	String- Introduction, methods, comprehension and its relative properties.	CLO04	B02, CH 8, Page No 207-235	Lecture	Test Questions	https://docs.python.org/3/tutorial/controlflow.html#documentation-strings	https://www.coursera.org/lecture/python-representation/python-strings- bbBTs https://www.coursera.org/lecture/python-data-analysis/python- more-on-strings-
27-28	Tuples, Set and Dictionaries- Introduction, methods, comprehension and its relative properties.	CLO04	B02, CH 10, Page No 307-318 B04, CH 09, Page No 257-281	Lecture	Test Questions	**https://docs.python.org/3/tutorial/datastructures.html** **https://docs.python.org/3/tutorial/datastructures.html#tuples-and-sequences**	https://nptel.ac.in/courses/106106182
29-32	Modules: Creating Modules, Variables in Modules, Imports and Attributes, Namespaces , Reloading, Built in modules, Generating Random values.	CLO05	B01, CH 5, Page No 101-111	Lecture	Brain Storming Sessions, Focused Listing	https://docs.python.org/3/tutorial/modules.html	https://www.coursera.org/lecture/python-genomics/lecture-6-modules-and- packages-10-32- rejwK
33-36	Files and Directories, File I/O, File positioning, File operators.	CLO05	B03, CH 7, Page No 289-307	Lecture	B rain Storming Sessions	https://python-course.eu/python-tutorial/file-management.php https://docs.python.org/3/tutorial/inputoutput.html#reading-and-writing-files	https://www.coursera.org/lecture/data-analytics-accountancy-1/python-file-i-o- cSSim https://www.coursera.org/lecture/accounting-data- analytics-python/5-1-python-file-io-R6L3h
37-40	Exception Handling: Introduction to Exception, Exception Handling, Except clause, Try, finally clause, User Defined Exceptions.	CLO05	B03, CH 12, Page No 480-494	Lecture	Leading Question	https://docs.python.org/3/tutorial/errors.html https://ehmatthes.github.io/pcc/cheatsheets/README.html#chapter-10-files-and-exceptions	https://www.coursera.org/projects/exception-handling-in-python
41-45	Exploring Object-Oriented Programming In Python- Introduction, Creating classes,	CLO06	B03, CH 09, Page No 400-426	Lecture	Leading Question, Quiz, One Minute Paper	https://docs.python.org/3/tutorial/classes.html?highlight=object%20oriented	https://www.coursera.org/learn/object-oriented-python

	objects, attributes. Constructors., Idea of inheritance, Superclass And SubClass. Overriding methods, Object class , Isinstance function.		B03, CH 10, Page No 436-447				

Lab Plan:

Lab No.	Experiment	CLO	TLM	ALM	Web References	Audio-Video
1-2	Anaconda Installation, Running Python Programs, Writing Python Scripts with Jupyter Notebook, and also make familiar with various platforms used for python programming, Implementation of basic concepts of python	CLO01	Demonstration method using a simulation or a tool	Quiz, Group Discussion	https://docs.anaconda.com/anaconda/navigator/tutorials/ https://www.anaconda.com	https://nptel.ac.in/courses/106106182
3-8	Data types(text type, numeric, sequence, mapping, set), Python basic operators- arithmetic, comparison, assignment operators, logical operators. , mix type conversion, integer to float and float to	CLO01	Discussion	Quiz, Student Created PPT, Flowcharts	https://www.learnpython.org/ https://www.geeksforgeeks.org/python-programming- language/	https://nptel.ac.in/courses/106106182
9-14	Coding problems related to conditional statements(if, elif, nested if else) and match statement.	CLO02	Demonstration method using a simulation or a tool	Quiz, Test Questions	https://www.learnpython.org/ https://docs.python.org/3/tutorial/controlflow.html	https://nptel.ac.in/courses/106106182
15-18	Coding problems related to for/while loops, For loop and range method, Break, continue and pass.	CLO02	Lecture	Test Questions	https://docs.python.org/3/tutorial/controlflow.html https://docs.python.org/3/tutorial/controlflow.html#for-	https://nptel.ac.in/courses/106106182
19-22	Nested loops and pattern designs	CLO02	Lecture	Test Questions	https://docs.python.org/3/tutorial/controlflow.html https://docs.python.org/3/tutorial/controlflow.html#for-statements	https://nptel.ac.in/courses/106106182
23-24	Implementation related to Functions: UserDefined and Built in functions(Python mathematical functions, random number functions), Passing Arguments(call by value/call by reference), Recursive Functions, Lambda Expressions.	CLO03	Lecture	Test Questions	https://docs.python.org/3/tutorial/controlflow.html#more-on-defining-functions	https://www.coursera.org/lecture/python-genomics/lecture-5-1-functions-part-1-5-54-PGXxZ https://www.coursera.org/lecture/python-genomics/lecture-5-2-functions-part-2-8-20-ylbSK
25-26	Implementation of lists. Creating empty list, Initializing list, list indexing and slicing operations, input method, Concatenating	CLO04	Lecture	Test Questions	https://docs.python.org/3/tutorial/datastructures.html	https://nptel.ac.in/courses/106106182

	multiple lists, generating range based lists. list based methods like append(), extend(), insert(), index(), count(), sort(), reverse(), pop() and nestedlists.					
27-28	Implementation of 2-D list with basic operations.	CLO04	Lecture	Brain Storming Sessions	https://docs.python.org/3/tutorial/datastructures.html	https://nptel.ac.in/courses/106106182
29	Python strings- accessing values in strings,updating strings, string special characters, string formatting operators, triple quotes, built in string methods. Capitalize(), count(), endswith, startswith, find, index, isalpha, isnum, isdigit, islower, isnumeric,isupper, lower, max, min, replace, splitswith, strip.	CLO04	Lecture	Brain Storming Sessions	https://docs.python.org/3/tutorial/controlflow.html#documentation-strings	https://www.coursera.org/lecture/python-representation/python-strings-bbBTs https://www.coursera.org/lecture/python
30	Implementation of Sets and relative methods.	CLO04	Lecture	Leading Question, Quiz	https://docs.python.org/3/tutorial/datastructures.html#tuples-and-sequences	https://nptel.ac.in/courses/106106182
31-32	Implementation of Tuples and relative methods.	CLO04	Lecture	Leading Question	https://docs.python.org/3/tutorial/datastructures.html#tuples-and-sequences	https://nptel.ac.in/courses/106106182
33-34	Creating empty dictionaries, initializing dictionaries, Accessing dictionary items, merging, and deletion. Understanding dictionary specific methods, keys (), values (), items (), copy (), update (), pop() and dictionary comprehension, Pythondate and time functions	CLO04	Lecture	Leading Question, Group Discussion	https://docs.python.org/3/tutorial/datastructures.html#tuples-and-sequences https://docs.python.org/3/tutorial/datastructures.html#sets https://docs.python.org/3/tutorial/datastructures.html#dictionaries	https://nptel.ac.in/courses/106106182
35	Python modules-import, from… importstatements, dir function, Generating Random values (random module)	CLO03	Lecture	Brain Storming Sessions	https://docs.python.org/3/tutorial/modules.html	https://www.coursera.org/lecture/python-genomics/lecture-6-modules-and-
36-37	File handling- Writing to a file, Reading from a file, Resetting the file currentposition, fseek	CLO05	Lecture	Brain Storming Sessions	https://python-course.eu/python-tutorial/file-management.php https://docs.python.org/3/tutorial/inputoutput.html#reading-and-writing-files	https://www.coursera.org/lecture/data-analytics-accountancy-1/python-file-i-o-cSSim https://www.coursera.org/lecture/accounting-data-analytics-python/5-1-python-file-io-R6L3h
38	Exploring Object-Oriented Programming In Python:, Creating classes, objects, attributes (implementation with Examples)	CLO06	Lecture and Inductive methods	Brain Storming Sessions	https://docs.python.org/3/tutorial/classes.html?highlight=object%20oriented	https://www.coursera.org/learn/object-oriented-python

39	Constructors and idea of inheritance., Superclass And SubClass (implementation with Examples)	CLO06	Lecture and Inductive methods	Brain Storming Sessions	https://docs.python.org/3/tutorial/classes.html?highlight=object%20oriented	https://www.coursera.org/learn/object-oriented-python
40-41	Overriding methods, Object class , Isinstance function, Exploring OperatorOverloading-Indexing And Slicing, Iteratable Objects (implementation with Examples)	CLO06	Lecture	Brain Storming Sessions	https://docs.python.org/3/tutorial/classes.html?highlight=object%20oriented	https://www.coursera.org/learn/object-oriented-python
42-45	Introduction to GUI Programming (Tkinter), Creating user defined modules,importing existing modules, Creating root windows and calling various dialog boxeso button click response.	CLO06	Lecture and Inductive methods	Brain Storming Sessions	https://docs.python.org/3/library/tkinter.html?highlight=tkinter#module-tkinter https://www.coursera.org/p rojects/build-a-python-gui- with-tkinter https://www.coursera.org/p rojects/gui-programming- login-system-python- tkinter	https://www.youtube.com/watch?v=YXPyB4XeYLA

8. Remedial Classes

After every Sessional Test, different types of learners will be identified and special discussions will be planned and scheduled accordingly.

Action Plan for different types of learners:

Learner Type-I	Learner Type- II	Learner Type- III
Remedial Classes, Doubt Sessions, Guided Tutorials	Workshop, Doubt Session	Coding Competitions, Project

9. Self-Learning

Assignments to promote self-learning, survey of contents from multiple sources.

S.No	Topics	CLO	ALM	References/MOOCS
1	Installation,Conditional Statements, Functions, Loops and Iterations	CLO01, CLO02, CLO05	Think – Pair- Share, Peer Review	Courseera-Python for Everybody Specialization(Getting Started with Python)
2	Data Types in Python, Loops, Strings, List	CLO01,CLO02, CLO04	Think – Pair- Share, Brain Storming sessions	Courseera- Crash Course on Python

10. Delivery Details of Content Beyond Syllabus

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

S.No	Advanced Topics, Additional Reading, Research papers and any	CLO	POs	ALM	References/MOOCS
1	Fundamentals of python library: Numpy : Array creation, Adding ,removing, and sorting elements of array, array dimension, reshaping, 1D, 2D arrays	CLO02, CLO03	PO1,PO2,PO3,PO4, PO5, PO12	Think – Pair- Share, Peer Review	https://numpy.org/doc/stable/user/basics.html https://numpy.org/doc/stable/user/absolute_beginners.html
2.	Fundamentals of python library: Pandas	CLO02, CLO03, CLO04	PO1,PO2,PO3,PO4, PO5, PO12	Think – Pair- Share, Peer Review, Brain storming	https://pandas.pydata.org/docs/user_guide/index.html

11. Evaluation Scheme & Components:

Assessment Type	Evaluation Component	Type of Component	No. of Assessments	% Weightage of Component	Max. Marks	Mode of Assessment	CLO
Formative	Component1	Continuous Lab Evaluations	01	15%	15	Offline	CLO01-CLO06
Summative	Component2	Sessional Tests(STs)	03	35%	35	Online/Offline	CLO01-CLO06
Summative	Component3	End Term Examination	01	50%	50	Offline	CLO01-CLO06
	Total		**100%**				

Note: Details in above table are sample for the reference, The department decides the assessment policy for each course.
All 3 STs are mandatory.

12. Syllabus of the Course:

Subject:			
S.No.	**Topic(s)**	**No. of Lectures**	**Weightage %**
1	Introduction to Python programming. Python environment setup, identifiers, reserved words ,lines and indentation, multiline statements, comments, quotation, input/output, output formatting(Integers and string). Fundamentals: Variables(simple, assigning multiple values, output variables, global/local variables), Python decision making- If else and Nested, If else and elif. Match statement, Introduction to while and for loop, Concept of nested looping. For loop and range method, Break, Continue and Pass, Pattern Design,	**14**	**31%**
2	Defining and calling Functions, Type of arguments (required, keyword, default, variable length), Scope Rules(Global/local Statements), Passing Arguments(call by value/call by reference), Recursive Functions, Lambda Expressions. Lists: Introduction to Lists, Operations on Lists, Storage Structure, Resizing and Looping on Lists, Indexing and Slicing, Passing variables through Functions, passing Lists through Functions, List Comprehension, map, filter & reduce functions, Two Dimensional Lists: Introduction and Storage, Jagged Lists, String- Introduction, methods, comprehension and its relative properties, Tuples, Set and Dictionaries- Introduction, methods, comprehension and its relative properties.	**14**	**31%**
3	Modules: Creating Modules, Variables in Modules, Imports and Attributes, Namespaces , Reloading, Built in modules, Generating Random values, Files and Directories, File I/O, File positioning, File operators, Exception Handling: Introduction to Exception, Exception Handling, Except clause, Try, finally clause, User Defined Exceptions.	**12**	**27%**
4	Exploring Object-Oriented Programming In Python-Introduction, Creating classes, objects, attributes. Constructors., Idea of inheritance, Superclass And SubClass. Overriding methods, Object class , Isinstance function.	**05**	**11%**

13. Academic Integrityy Policy:

Education at Chitkara University builds on the principle that excellence requires freedom where Honesty and integrity are its prerequisites. Academic honesty in the advancement of knowledge requires that all students and Faculty respect the integrity of one another's work and recognize the importance of acknowledging and safeguarding intellectual property. Any breach of the same will be tantamount to severe academic penalties.

This Document is approved by:

Designation	**Name**	**Signature**
Course Coordinator	Dr. Mandeep Kaur	
Head-Academic Delivery	Dr. Renu Popli	

Assistant Dean	Dr. Deepak Thakur	
Date(DD/MM/YYYY)	14/08/2024	